

Random Variable Transformation Techniques CPE 486/586

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1 Function of a random variable

Assume that X is a random variable with pmf/pdf f_X , cdf F_X , and the sample space $\mathcal{X}$.

Let $Y = g(X)$, e.g., $Y = X$, $X + 5$, $3X^2$, e^{-X} , $|X|$, $I\{X > 0\}$. Then

- Y is a random variable in its own right.

Question: How is the distribution of Y related to that of X ?

Denote the sample space of Y by $\mathcal{Y}$. For any set $A \subset \mathcal{Y}$, we have

$$P(Y \in A) = P(g(X) \in A) \quad (1)$$

$$= P(X \in g^{-1}(A)), \quad (2)$$

where

$$g^{-1}(A) = \{x : g(x) \in A\}.$$

Note that

- $g^{-1}(A)$ is an inverse mapping, which is defined for any g (g is not necessarily one-to-one and/or onto).
- $g^{-1}(A)$ is a mapping from subsets of $\mathcal{Y}$ to subsets of $\mathcal{X}$.

2 Transformation of Discrete X

Assume X is discrete with the pmf $f_X(x) = P(X = x)$.

Let $Y = g(X)$, then the sample space of Y is

$$\mathcal{Y} = \{y : y = g(x), \text{ for some } x \in \mathcal{X}\}.$$

Y is also discrete.

The pmf of Y can be computed as following

$$f_Y(y) = P(Y = y) \quad (3)$$

$$= P(g(X) = y) \quad (4)$$

$$= P(X \in g^{-1}(\{y\})) \quad (5)$$

$$= \sum_{x:g(x)=y} f_X(x), \quad \text{for any } y \in \mathcal{Y}. \quad (6)$$

Example 1. The distribution of X is

x	-2	-1	0	1	2
$f_X(x)$	0.1	0.2	0.4	0.2	0.1

$Y = |X|$. Find the pmf of Y .

Example 2. Toss a coin n times whose probability of head is p . X is the number of heads. Then X has a binomial distribution, denoted as $\sim \text{Bin}(n, p)$, with the pmf

$$f_X(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, \dots, n.$$

Let Y denote the number of tails, i.e., $Y = n - X$. Find the pmf of Y .

3 Transformation of Continuous X

It is convenient to use

$$\mathcal{X} = \{x : f_X(x) > 0\}, \quad \mathcal{Y} = \{y : y = g(x) \text{ for some } x \in \mathcal{X}\}.$$

The set $\{x : f_X(x) > 0\}$ is the support set of X . The cdf of $Y = g(X)$ is

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) \quad (7)$$

$$= P(\{x \in \mathcal{X} : g(x) \leq y\}) \quad (8)$$

$$= \int_{\{x \in \mathcal{X} : g(x) \leq y\}} f_X(x) dx. \quad (9)$$

3.1 g Strictly Monotone (Increasing or Decreasing)

Theorem 1. Let $X \sim F_X$ with support $\mathcal{X}$. Let $Y = g(X) \sim F_Y$ with the sample space $\mathcal{Y}$.

- (a) If g is increasing, then $F_Y(y) = F_X(g^{-1}(y))$.
- (b) If g is decreasing, then $F_Y(y) = 1 - F_X(g^{-1}(y))$.
- (c) Assume X has density f_X . If g is monotone and g^{-1} is continuously differentiable on $\mathcal{Y}$. Then

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|, & y \in \mathcal{Y}, \\ 0, & \text{otherwise.} \end{cases}$$

Example 3. $X \sim \text{Unif}[0, 1]$.

- $Y = X^3$
- $Y = 1/X$
- $Y = -\log X$ (*uniform-exponential*)

Theorem 2 (cdf Transformation). Let X have continuous cdf $F_X(x)$. Define the cdf transformation $Y = F_X(X)$. Then $Y \sim \text{Unif}[0, 1]$.

Application: A general method to generating random variables.

Let F be any cdf. If you want to generate an observation from a population with cdf F , do the following

- (i) Generate a uniform random number u from $[0, 1]$, i.e. $U \sim \text{Unif}[0, 1]$.
- (ii) If F is strictly monotone, define $x = F^{-1}(u)$; otherwise define x as $\inf\{x : F(x) \geq u\}$.

Then $X \sim F$.

This is called **Inverse Transform Sampling**.

Example 4. *Exponential* $F(x) = 1 - e^{-x/\beta}$, $x > 0$.

Example 5. *Logistic* $F(x) = \frac{e^x}{1+e^x}$, $x > 0$.

3.2 g Piecewise Monotone

Example 6. X has cdf F_X (pdf f_X) on $\mathbb{R}$. $Y = X^2$.

Note $\mathcal{Y} = [0, \infty)$. For any $y \geq 0$, we have

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) \quad (10)$$

$$= P(-\sqrt{y} \leq X \leq \sqrt{y}) \quad (11)$$

$$= P(X \leq \sqrt{y}) - P(X \leq -\sqrt{y}) \quad (12)$$

$$= F_X(\sqrt{y}) - F_X(-\sqrt{y}). \quad (13)$$

Differentiating $F_Y(y)$ with respect to y ,

$$f_Y(y) = (d/dy)F_Y(y) \quad (14)$$

$$= f_X(\sqrt{y}) \frac{1}{2\sqrt{y}} - f_X(-\sqrt{y}) \left(-\frac{1}{2\sqrt{y}}\right) \quad (15)$$

$$= \frac{f_X(\sqrt{y}) + f_X(-\sqrt{y})}{2\sqrt{y}}. \quad (16)$$

Theorem 3. Let $X \sim f_X(x)$, $x \in \mathcal{X}$ and $Y = g(X)$. Assume $\mathcal{X}$ has a partition

$$\mathcal{X} = A_0 \cup A_1 \cup \dots \cup A_k,$$

which satisfies that

- (i) $P(X \in A_0) = 0$, exceptional set
- (ii) f_X is continuous on each A_i , $i = 0, \dots, k$,
- (iii) $g_i = g|_{A_i}$ is monotone for each A_i , $i = 1, \dots, k$,
- (iv) Range of g_i is the same $\mathcal{Y}$ for each $i = 1, \dots, k$,
- (v) $g_i^{-1}(y)$ has a continuous derivative on $\mathcal{Y}$ for each $i = 1, \dots, k$.

Then

$$f_Y(y) = \begin{cases} \sum_{i=1}^k f_X(g_i^{-1}(y)) \left| \frac{d}{dy}(g_i^{-1}(y)) \right|, & y \in \mathcal{Y}, \\ 0, & \text{otherwise.} \end{cases}$$

Example 7. Assume X has the pdf $f_X(x) = (2\pi)^{-1/2}e^{-x^2/2}$, $x \in \mathbb{R}$ (standard Normal, denoted as $N(0,1)$).

- Let $Y = X^2$

- Let $Y = X^4$.

Example 8. Assume $X \sim \text{Uniform}[-1, 1]$.

Let $Y = X^2$ for $X \leq 0$ and $Y = X$ for $X > 0$.

Example 9. Assume $X \sim N(0, 1)$.

Let $Y = X^2$ for $X \leq 0$ and $Y = X$ for $X > 0$.

3.3 More Complicated Transformations

Example 10. Let $X \sim \text{Unif}[-1, 2]$, i.e. its pdf is given by

$$f_X(x) = \begin{cases} \frac{1}{3}, & \text{if } x \in [-1, 2] \\ 0 & \text{otherwise.} \end{cases}$$

Let $Y = X^2$.

4 Expected Values

X random variable with pdf or pmf $f(x)$.

The expected value or mean of $g(X)$ is defined as

$$E(g(X)) = \begin{cases} \int_{-\infty}^{\infty} g(x)f(x) dx, & \text{if } X \text{ is continuous,} \\ \sum_{x \in \mathcal{X}} g(x)f(x), & \text{if } X \text{ is discrete.} \end{cases}$$

provided that $E|g(X)| < \infty$. If $E|g(X)| = \infty$, we say that the expectation does not exist.

Example 11. $X \sim \text{Unif}[0, 1]$. Find $E(X)$ and $E(X^2)$.

Example 12. X is Binomial(n, p) with the pmf

$$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad \text{for } x = 0, \dots, n.$$

Example 13. X is Exp(β) with the pdf

$$f_X(x) = \beta^{-1} e^{-x/\beta}, \quad \text{for } x > 0; \quad = 0 \text{ otherwise.}$$

Find $E(X)$ and $E(e^X)$.

Example 14 (Cauchy). X has the pdf $f_X(x) = \pi^{-1}(1+x^2)^{-1}$ for all x .

More examples will be given in the next section.

Theorem 4. Assume $Eg_1(X)$ and $Eg_2(X)$ exist.

1. $E(ag_1(X) + bg_2(X) + c) = aE(g_1(X)) + bE(g_2(X)) + c$.
2. $g(x) \geq 0 \Rightarrow E(g(X)) \geq 0$.
3. $g_1(x) \geq g_2(x) \Rightarrow E(g_1(X)) \geq E(g_2(X))$.
4. $a \leq g(x) \leq b \Rightarrow a \leq E(g(X)) \leq b$.
5. $E(X - E(X)) = 0$.
6. $E(X - b)^2$ is minimized for $b = E(X)$.

Remark 1 (Two Ways of Computing Expected Values). Let $X \sim f_X$, $Y = g(X)$, $Y \sim f_Y$.

$$E(g(X)) = \int g(x)f_X(x) dx. \quad (17)$$

$$E(Y) = \int yf_Y(y) dy. \quad (18)$$

Are they equal?

Example 15. $X \sim \text{Unif}(0,1)$. Let $g(X) = -\log(X)$.

Remark 2 (Expectation of Non-negative Random Variables). Assume $X \geq 0$ and has the cdf F_X . Then

$$E(X) = \begin{cases} \int_0^\infty (1 - F_X(x)) dx, & X \text{ continuous,} \\ \sum_{k=0}^\infty (1 - F_X(k)), & X \text{ discrete.} \end{cases}$$

Example 16. Mean of exponential.

5 Median

Assume X is continuous and has the cdf F_X . Its median m is the value which satisfies $F_X(m) = 1/2$, that is,

$$\int_{-\infty}^m f_X(x) dx = \int_m^\infty f_X(x) dx = \frac{1}{2}.$$

Equivalently, $m = F_X^{-1}(1/2)$.

Example 17. Assume X has the pdf $f(x) = 3x^2$, $0 < x < 1$.

Example 18. Assume $X \sim \text{Exponential}(\beta)$.

Example 19. Cauchy.

Example 20. Symmetric.

6 Variance and Standard Deviation

The variance of a random variable X is defined as

$$\text{var}(X) = E(X - E(X))^2.$$

Use $\sigma^2 = \text{var}(X)$. The standard deviation of X is the square root of $\text{var}(X)$

$$\sigma = \sqrt{\text{var}(X)}.$$

- Both variance and standard deviation measure the degree of spread of a distribution around its mean $E(X)$.

Theorem 5 (Computational formula).

$$\text{var}(X) = E(X^2) - (E(X))^2, \quad (19)$$

$$\text{var}(aX + b) = a^2 \text{var}(X). \quad (20)$$

Example 21. $X \sim \text{Unif}[0, 1]$. Compute $\text{var}(X)$.

Example 22 (Binomial). Assume X is discrete and has the pmf

$$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad \text{for } x = 0, \dots, n.$$

7 Moments

For each positive integer r , the r th (raw) moment of X is

$$\mu'_r = E(X^r).$$

The r th central moment

$$\mu_r = E(X - E(X))^r.$$

Note that

$$\mu'_0 = \mu_0 = 1, \quad \mu'_1 = E(X), \quad \mu_2 = \text{var}(X).$$

Example 23. *Binomial.*

The r th factorial moment of X is defined as

$$\mu^{[r]} = E(X(X-1) \cdots (X-r+1)).$$

It is useful for the calculation of moments of discrete distributions.

7.1 Two special moments: Skewness and Kurtosis

Skewness of a distribution measures its departure from symmetry

$$\gamma_1 = \mu_3 / \sigma^3 = \mu_3 / \mu_2^{3/2}.$$

- For symmetric random variables, $\mu_3 = E(X - E(X))^3 = 0$, so is γ_1 .
- γ_1 is not unit free.

Example 24. *Exponential.*

Kurtosis of a distribution measures its peakedness of the distribution, defined using the fourth moment

$$\gamma_2 = \frac{\mu_4}{\mu_2^2} - 3 = \frac{\mu_4}{\sigma^4} - 3.$$

- Division by σ^4 is to make kurtosis a pure number.
- Subtraction of 3 is a convention, so that kurtosis is zero for the normal distribution.
- $\gamma_2 > 0$ leptokurtic, $\gamma_2 < 0$ platykurtic.
- For the normal, $\gamma_1 = \gamma_2 = 0$.

Example 25. *Uniform.*

Remark 3. • Mean, variance, skewness, kurtosis are good summary measures for a distribution. However, they do not characterize a distribution.

- In fact, even if we have all moments equal, we may not end up with the same distribution.

Example 26 (Moments do not uniquely define a distribution). Suppose that Y has density $f_Y(y) = (2\pi)^{-1/2}e^{-y^2/2}$ for all y (standard Normal).

- (1) Let $X_1 = e^Y$. What is the density $f_1(x)$ of X ?
- (2) Define $f_2(x) = f_1(x)(1 + \sin(2\pi \log x))$, $x > 0$. Prove that f_2 is a pdf.
- (3) Suppose $X_2 \sim f_2(x)$. Show that X_1 and X_2 have the same moments for all r , i.e.,

$$\int_0^\infty x^r f_1(x) dx = \int_0^\infty x^r f_2(x) dx, \quad \forall r = 0, 1, \dots$$

8 Moment Generating Function

Assume X has the cdf F_X and the pmf/pdf f_X . The moment generating function (MGF) of X is defined as

$$M_X(t) = E(e^{tX}) = \begin{cases} \sum e^{tx} f_X(x), & \text{if } X \text{ discrete,} \\ \int_{-\infty}^\infty e^{tx} f_X(x) dx, & \text{if } X \text{ continuous.} \end{cases}$$

- **Interpretation:** Note

$$e^{tX} = 1 + \frac{tX}{1!} + \frac{t^2 X^2}{2!} + \dots$$

so intuitively

$$E(e^{tX}) = 1 + \frac{tE(X)}{1!} + \frac{t^2 E(X^2)}{2!} + \dots$$

Thus, the coefficient of $t^r/r!$ in the infinite Taylor series expansion of $M_X(t)$ is $\mu'_r = E(X^r)$. That's why $M(t)$ is called the moment generating function — its expansion generates the moments.

- In the same way, μ_r is obtained from the expansion of $E(e^{t(X-E(X))}) = e^{-tE(X)} M_X(t)$.
- Note that the above step needs some justification, as we know that the expectation of a sum is equal to the sum of expectations only when there are finitely many terms involved. This is justified whenever $M(t) < \infty$ for all t in an open interval containing 0.

8.1 Computation Formula

$$M(0) = 1. \quad (21)$$

$$M_{aX+b}(t) = e^{tb} M_X(at). \quad (22)$$

$$\mu'_r = E(X^r) = \left(\frac{d^r}{dt^r} M_X(t) \right)_{t=0}. \quad (23)$$

Example 27. *Binomial.*

Example 28. *Exponential.*

Example 29. *Uniform*

8.2 Gamma function

An expression that often appears is an integral of the form $\int_0^\infty e^{-x} x^{\alpha-1} dx$, where $\alpha > 0$. It can be shown that for all $\alpha > 0$, the above integral is finite. Obviously, its value depends on α . We denote the value by $\Gamma(\alpha)$, i.e.,

$$\Gamma(\alpha) = \int_0^\infty e^{-x} x^{\alpha-1} dx,$$

called the gamma function. Note that

- $\Gamma(1) = \int_0^\infty e^{-x} dx = 1.$
- If $\alpha > 1$, then by integration by parts,

$$\Gamma(\alpha + 1) = [x^\alpha (-e^{-x})]_0^\infty - \int_0^\infty (-e^{-x}) \alpha x^{\alpha-1} dx = \alpha \Gamma(\alpha).$$

- In particular, for any positive integer $n > 1$,

$$\Gamma(n) = (n-1)\Gamma(n-1) = (n-1)(n-2)\Gamma(n-2) = \cdots = (n-1)(n-2) \cdots 1\Gamma(1) = (n-1)!.$$

8.3 What determines the distribution uniquely?

- If X and Y have bounded supports, and share the same moment sequence, then they have the same distribution.
- If $M_X(t) = M_Y(t) < \infty$ in a neighborhood of 0, then X and Y have the same distribution.

The above also holds in a limiting sense.

- If X_n is a sequence of random variables and X is another such that

$$M_{X_n}(t) \rightarrow M_X(t), \quad \forall t \text{ in a neighborhood of } 0,$$

then the distribution of X_n “converges” to the distribution of X . We say that X_n converges to X in distribution.

(If X is continuous, the “convergence” means that $F_{X_n}(x) \rightarrow F_X(x)$ for all continuous x .)

(If X_n and X are all discrete taking values on $\{0, 1, \dots\}$, then “convergence” means convergence of pmf’s.)

Example 30 (Convergence of binomial to Poisson). X the number of successes in n trials with probability of success p .

n large, p small, but np is moderate.

$n \rightarrow \infty$, $p \rightarrow 0$, $np \rightarrow \lambda$.